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State of Charge Estimation of Lithium-Ion Batteries Based on Temporal Convolutional Network and Transfer Learning

YUEFENG LIU[1], JIAQI LI[1], GONG ZHANG[1], BIN HUA[2], AND NEAL XIONG[2,3], (Senior Member, IEEE)

[1]School of Information Engineering, Inner Mongolia University of Science and Technology, Baotou 014010, China
[2]School of Information Engineering, Ningxia University, Yinchuan 750000, China
[3]Department of Mathematics and Computer Science, Northeastern State University, Tahlequah, OK 74464, USA

Corresponding author: Bin Hua (huabin@nxu.edu.cn)

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ABSTRACT Accurate estimation of the state of charge (SOC) is critical for the normal use of lithium-ion battery equipment like electric vehicles. However, the SOC of lithium-ion battery is not available by direct measure, but can only indirectly be estimated by measurable variables. According to the nonlinear characteristics between the measured values and SOC during the working period of lithium-ion batteries, we propose a method to estimate the SOC of lithium-ion batteries with Temporal Convolutional Network (TCN). The measured values of voltage, current, and temperature during the use of lithium-ion batteries can be directly mapped to accurate SOC in this method without using a battery model or adaptive filter. The network can self-learning and update parameters by being fed datasets collected under various working conditions and then obtain a model that can correctly estimate SOC under different estimation conditions. In addition, it can also be applied to different types of lithium-ion batteries through transfer learning with only a small amount of battery data. At various ambient temperature conditions, the average MAE estimated by the proposed method is 0.67% for all the tests, which proves that the TCN network is an effective tool to estimate the SOC of lithium-ion batteries.

INDEX TERMS Deep learning, lithium-ion batteries, state of charge estimation, temporal convolutional network (TCN), transfer learning.

I. INTRODUCTION

The world is undergoing an energy transition driven both by improving climate change and promoting sustainable growth. As renewable energy technologies continue to evolve, the power sector has seen impressive growth. When electricity comes from renewable energy, it becomes a kind of clean energy. Countries are also more inclined to take electricity as the preferred ultimate energy carrier in energy transformation. Electrification is becoming a key solution for reducing emissions [1]. Electric vehicles (EV) have grown rapidly on the back of national policies, with their zero emissions, greater efficiency than internal combustion engines, and huge potential to reduce greenhouse gas emissions when combined with low-carbon electricity [2]. In recent years, lithium-ion battery as one of the great prospects for the development of energy storage technology has been widely attention, since it has high energy density, low self-discharge performance, close to zero memory effect, high open-circuit voltage, and long service life, the lithium-ion batteries with high energy density are considered to be the ideal power source for the EV industry [3].

State of Charge (SOC), defined as the ratio of battery residual capacity to nominal capacity, is one of the important characteristics of a battery that can be used to describe the current state of the battery in use. Accurate SOC can protect the battery from overcharge or discharge and it is an essential part of the battery management system (BMS) [4]. Due to the SOC of the battery can only be estimated indirectly through the observable variables of the battery, such as voltage, current, and temperature, and may be affected by the battery aging, environmental temperature changes, various vehicle driving conditions, and other uncertainties. As a consequence, it is an complex and challenging task to estimate the SOC accurately [5].

Traditionally, there are two methods to estimate SOC, the open-circuit techniques [6] and the Coulomb counting methods [7]. As is well-known, these methods have

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certain limitations. For example, the open-circuit voltage techniques need to maintain a static state for a long time to obtain the SOC curve corresponding to the open circuit voltage (OCV) [8], while the Coulomb counting methods need to obtain the initial SOC with the accumulated error [9]. More complex methods, such as adaptive filtering algorithms, are usually used to estimate the battery SOC based on the current and voltage. For example, the Kalman Filtering algorithm and its variants [10]–[12], Particle Filter algorithm [13], [14], etc. In the paper [11], a simplified equivalent circuit model was established to simulate the dynamic characteristics of the battery, in which OCV was used as the linearization function of SOC, and then the system state was estimated according to the Kalman Filtering algorithm. Finally, the effectiveness of the method was verified through experiments on $LiFePO_4$ lithium-ion battery under different operating currents and temperatures. In the paper [13], an online estimation method of SOC based on adaptive particle filter was proposed. Based on the traditional adaptive particle swarm algorithm, there was an error analysis method was added to the adaptive particle filter algorithm, so as to improve the robustness. The experimental analysis showed that the maximum error of both voltage and SOC estimation were less than 1%. However, these methods also have some limitations that require additional parameters or establish a reliable battery model to improve the accuracy and precision of estimation [4].

With the development of artificial intelligence technology, applications in various fields have gradually matured [15]–[18]. Deep learning methods have also been introduced to the SOC estimation of lithium-ion batteries [19]–[26].

Since the battery data is time series data, the common method of processing time series data in the industry is to use the Long Short-Term Memory (LSTM) network which is a variant of Recurrent Neural Network (RNN). In the paper [21], the author used the LSTM-RNN model to estimate the SOC of the battery, which avoided the problems of vanishing or exploding gradient in the backpropagation of RNN and achieved better success in capturing the long-term dependence relationship within the sequence. Through the author's verification, the model can directly map the measurements of the battery voltage, current, and temperature to the battery SOC, and obtain great performance. In the paper [24], the Gated Recurrent Unit (GRU) network, a variant of the LSTM network, was adopted to capture the mapping relationship between the SOC and measurements of the battery including voltage, current, and temperature, and achieved outstanding performance in both speed and accuracy. LSTM has been playing an important role for a long time. However, due to the long-time span of battery data, and the RNN structure can only read and process one time moment at a time, the long input information will lead to the increase of computation and training time and difficulty.

Lea *et al.* proposed a time convolutional network for video-based action segmentation in their groundbreaking work in 2016 [27]. This traditional process consists of two parts. Firstly, the Convolutional Neural Network (CNN) is used to calculate low-level features and then RNN is used to input these low-level features into a classifier to obtain high-level time information. The main drawback of this approach is that it requires two separate models. Bai *et al.* put forward a general TCN model in 2018 [28], which provides a unified approach to capture in the form of layers all two levels of information. It combines the advantage of CNN and RNN, in which, the causal convolution makes forecasts of future data don't leak, and the dilated causal convolution and residual structure can achieve a larger receptive field within a finite layer, and the calculation amount and training time of the model has no obvious increase. Relevant researchers soon applied TCN to time series forecasting problems, such as power timing prediction [29], traffic flow prediction [30], [31], and bearing residual service life prediction [32]. The experimental results show the effectiveness of TCN.

At the same time, neural network model training always relies on numbers of training data. The data required by the training model must cover as much operation range as possible required by the application scene, and the trained model is only targeted at specific battery types. In the paper [22], the transfer learning technique was applied to the SOC estimation of lithium-ion battery. Based on the LSTM neural network, three kinds of models were implemented for SOC estimation for four different lithium-ion batteries, including models trained traditionally with no transfer learning, models trained with transfer learning using full target dataset, and models trained with transfer learning using partial target dataset. And their performance was also compared. Results showed the prospect of transfer learning in reducing training time, improving SOC estimation accuracy, and reducing the amount of training data required.

To our knowledge, this is the first time for TCN to be applied in the SOC estimation of lithium-ion batteries and the first time for transfer learning to be used on the TCN network. In this paper, we demonstrated how the TCN network applied to estimate SOC of lithium-ion batteries and validated its estimation effect widely. Then the feasibility of this method combined with transfer learning was also validated. Compared with some recently proposed methods, the comparison of this method also shows certain advantages. Briefly speaking, the main contributions of this paper are as follows:

1) With its special structure, TCN can capture the measurements of the battery directly map to the SOC, avoiding intensive calculation and more suitable for the on-board system.
2) TCN can learn by itself and update the parameters using the backpropagation algorithm, without manual design and parametric battery model.
3) The TCN model can estimate the SOC of the battery under various working conditions, without using different models or different lookup tables.
4) The trained TCN model can be applied to other types of battery SOC through transfer learning technology,

which reduces the cost of model design and large dataset establishment.

In the rest of the paper, the related work is introduced in Section II. In Section III the details of the proposed methods are given. Section IV describes the experiments and verifies the estimated results of the methods in varied conditions. Finally, the conclusions are drawn in Section V.

II. RELATED WORK

The advantage of deep learning technology in capturing nonlinear relationships makes it show good performance and robustness in estimating lithium-ion battery SOC. In recent years, with the development and maturity of deep learning, new networks were constantly proposed, and neural network models were also improving in speed and data processing capacity. Meanwhile, the development of transfer learning technology has greatly promoted the application of deep learning technology in various fields. In this section, we will briefly introduce the related work about recurrent neural network, time convolutional network, and transfer learning.

A. RECURRENT NEURAL NETWORK

Similar to an ordinary neural network, RNN has an input layer, an output layer, and hidden layers. What's different is that RNN has different states at different time t, in which the output of the hidden layer at time t-1 will act on the hidden layer at time t. So RNN can capture the dependencies in the sequential data.

The Back Propagation Through Time (BPTT) algorithm is used in the RNN training process. Sometimes, it is prone to have vanishing or exploding gradient so that it can't solve the long-term dependence problem well. Then, the emergence of LSTM solves this problem [33]. LSTM has the same basic structure as RNN, but the internal structure of its loop module is specially designed to deal with long-term dependencies. The Long Short-Term Memory cell is equipped with a cell state which stores the long-term dependencies and during training, the input, output and forget gates allow the LSTM to forget or write new information to the memory cell.

Then, on the basis of LSTM, GRU, a variant of LSTM, appeared [34]. It combines forget gate and input gate into a single ''update gate'', merges Cell State and hidden State, and makes some other changes to make the model simpler than the standard LSTM model.

B. TEMPORAL CONVOLUTIONAL NETWORK

TCN is a special kind of convolutional neural network, which can carry out large-scale parallel processing and shorten the time of network training and verification. At the same time, by stacking more convolutional layers, using the greater dilated rate of expansion, and increasing the filter size, TCN provides more flexibility to change the receptive field size and can better control the memory length of the model. The path of backpropagation and time direction of the sequence of TCN are different, which avoids the gradient explosion or gradient disappearance problem frequently occurring in RNN. In addition, it requires less memory for training, especially for long input sequences. Specifically, these advantages of TCN are attributed to the design of its dilated causal convolution and the introduction of the residual model.

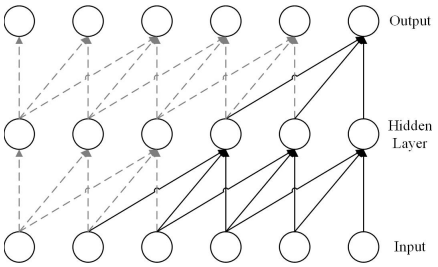


FIGURE 1. Causal convolution block with two layers with kernel size 3.

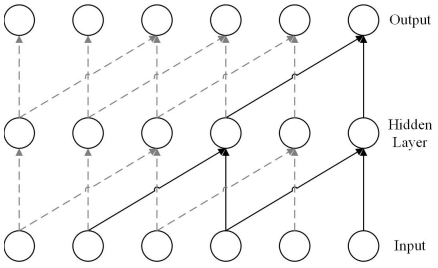


FIGURE 2. Dilated causal convolution block with two layers with kernel size 2, dilation rate 2.

1) DILATED CAUSAL CONVOLUTIONS

To ensure the input sequence and the output sequence of each hidden layer are the same length, one-dimensional convolution with zero-padding is adopted in the TCN network, so that the network can be adjusted more flexibly according to the needs. The data at time t can only be obtained by convolving the data of the previous layer at time t and before, which effectively prevents the leakage of future data. Fig.1 shows a causal convolution block.

To increase the receptive field without using pooling operations, holes can also be injected into the convolution. This means a fixed step is introduced between each two adjacent filter taps. For a one-dimensional input sequence $x \in X$ and filter $f : \{0, \cdots, k-1\} \rightarrow \mathbb{R}$, the dilated convolution operation F of sequence element s is defined as:

$$F(s) = (x *_d f)(s) = \sum_{i=0}^{k=1} f(i) \cdot x_{s-d \cdot i} \tag{1}$$

where d is the dilate factor, k is the filter size, and $s - d \cdot i$ represents the past direction. Specially in the case of $d = 1$, the dilated convolution will be equivalent to regular convolution. The dilated causal convolution can help TCN to capturing longer-term patterns. Fig.2 shows a dilated causal convolution block.

2) RESIDUAL BLOCK

The network depth, filter size, and dilation factor determine the size of the receptive field of the TCN model. In order to obtain a large receptive field, the depth of the network

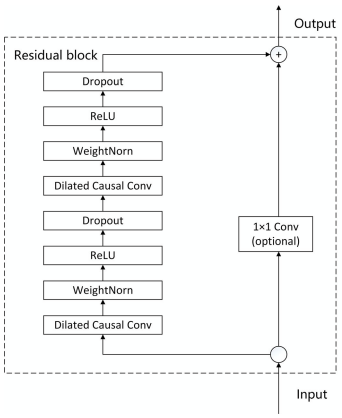


FIGURE 3. TCN residual block which has two round of dilated causal convolution, weight normalization, activation function, and dropout. And an 1 × 1 convolution is added when residual input and output have different dimensions.

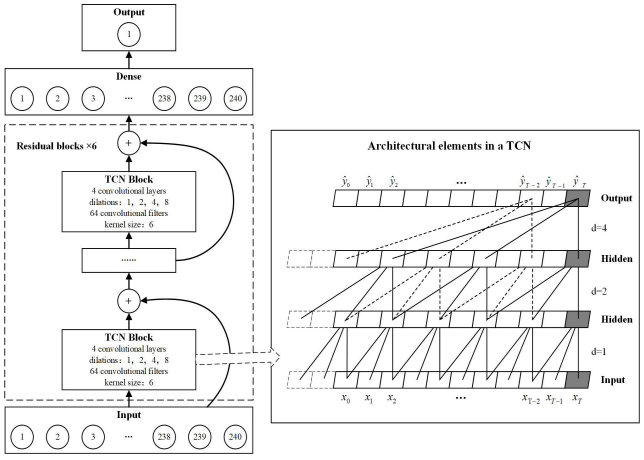


FIGURE 4. The proposed TCN model for SOC estimation. The model has six stacks, each stack has four convolutional layers of kernel size 6, 64 filters, and the dilations are 1, 2, 4, 8 respectively.

needs to be increased, therefore it also adopts the residual connection to help build a deeper and larger model. The residual structure can help the deep network avoid the vanishing or exploding gradient problem, which is beneficial to a very deep network.

TCN uses a general residual module in place of its convolutional layer. The input of the residual block for TCN goes through two rounds of dilated causal convolution, weight normalization, rectified linear unit (ReLU) activation function, and dropout, then a convolution ensures that the output dimension is the same as the input. Fig. 3 shows the residual block of TCN.

C. TRANSFER LEARNING

Different from the traditional training method of neural network model, which requires training from scratch. Considering the correlation of many tasks or data, the model parameters that have been trained on a certain task or data can be shared with the new model in a certain way, so as to help the new model train better [35]. In addition, transfer learning can also reduce the dependence of training model on numbers of training data, thus reducing the training cost.

The mathematics of transfer learning is defined as: Given a source domain $D_s = \{X_s, f_S(X)\}$ and learning task T_S, a target domain $D_T = \{X_T, f_T(X)\}$ and learning task T_T, the aim of transfer learning is to help improve the learning of the target predictive function $f_T(\cdot)$ in D_T using the knowledge in D_S and T_S, where $D_S \neq D_T$, or $T_S \neq T_T$.

III. OUR PROPOSED METHODS FOR SOC ESTIMATION

A. ESTIMATION BASED ON THE PROPOSED TCN NETWORK

Our proposed network starts with a sequence input layer, where the input vector is the measurements including voltage, current, and temperature. Then, the TCN layer contained six stacks, each of which has four convolutional layers of kernel size 6, 64 filters, and dilations of 1, 2, 4, 8 are followed to learn the dependence on input. Meanwhile, in the TCN layer, the residual module including a 1D convolution is employed

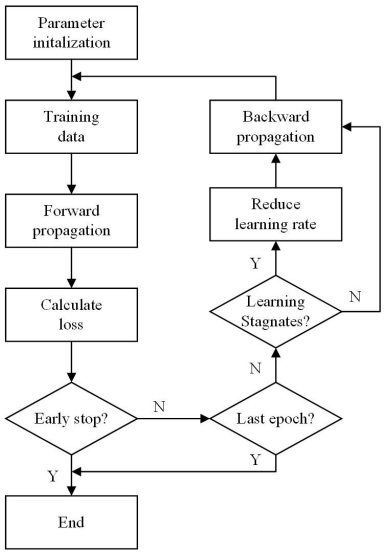


FIGURE 5. The training process, in which a learning rate decline strategy and an early stop mechanism are introduced.

to replace the convolutional layer. Finally, a fully connected layer is used to estimate the SOC.

To extract more advanced features, the input sequence is processed by the dilated causal convolution, and identity mappings are also introduced to help training simultaneously. The architecture of the proposed TCN model for SOC estimation is represented in Fig. 4.

TCN estimation consists of two parts: offline training and online estimation. During the offline training process, a large amount of battery data is fed to the network to let the network capture the nonlinear relationship between the battery measurements and the corresponding SOC. To speed up the training process as well as avoid the occurrence of overfitting, the learning rate decline strategy, and the early stop mechanism are also introduced. The training process is depicted in Fig. 5. When the offline training process is complete, the trained model will be able to be applied to online estimation.

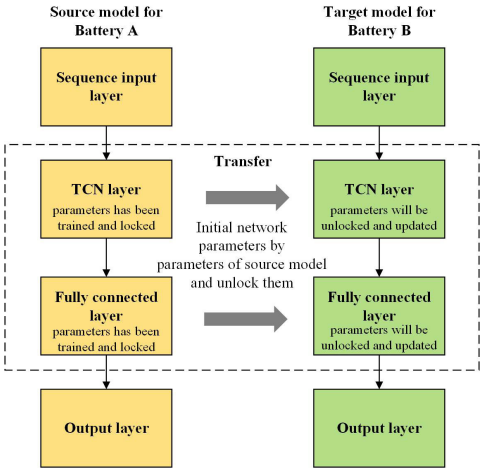


FIGURE 6. The transfer learning. The two models have the same structure, the target model is initialized with the parameters of the source model, and then the new target dataset is used for training and the parameters are updated in the process.

B. ESTIMATION BASED ON THE TRANSFER LEARNING

Transfer learning is the transfer of a model trained on a previous learning task to a new task. The spatial characteristics of different measurement parameters of lithium-ion batteries are different but related. Therefore, we first train and apply a TCN model for battery SOC estimation, and then use this model as a pre-training model to train another model for battery SOC estimation. During transfer learning, network parameters are unlocked to allow them to be updated to new values during training.

Fig. 6 depicts the transfer learning approach proposed in this paper.

IV. EXPERIMENTS AND RESULTS ANALYSIS

In this chapter, we first described the experimental setting and introduced the data set used in the experiment. Then, we expound on a series of experiments to extensively verify the proposed model, including the estimation of the proposed method under fixed temperature, multi-temperature, and varying temperature environments as well as the analysis of the experimental results. At the same time, the trained model was also applied to estimate the SOC of a different kind of lithium-ion battery by transfer learning technology, and its effect was tested. Finally, the proposed TCN model in this paper was compared with the model based on LSTM-RNN and GRU-RNN.

In this paper, TensorFlow Keras was used in conjunction with an NVIDIA Tesla V100 GPU with 32GB memory to implement and train the model.

A. EXPERMENT SETTING

The input vector of the TCN model is given by $x(k) = [V(k), I(k), T(k)]$, where $V(k)$, $I(k)$, and $T(k)$ represent the measured voltage, current, and temperature of the battery at time step k, respectively. The output of the TCN model is the estimated SOC at the same time step, $y(k) = [SOC(k)]$. The batch size and the number of epochs of training are 32 and 100 respectively.

TABLE 1. Panasonic 18650PF battery main specifications.

Chemistry	LiNiCoAlO2 or NCA
Nominal Open Circuit Voltage	3.6V
Capacity	Min. 2.75 Ah / Typ. 2.9 Ah
Min / Max Voltage	2.5V / 4.2V
Mass / Energy Storage	48g / 9.9Wh
Minimum Charging Temperature	10°C
Cycles to 80% Capacity	500 (100% DOD, 25°C)

In order to facilitate the model training, we normalized data to the range [-1,1] as shown in formula (2):

$$x_{norm} = \frac{2(x - x_{\min})}{x_{\max} - x_{\min}} - 1 \tag{2}$$

where x_{max} and x_{min} are the maximum and the minimum values of data, x represents the initial data, and x_{norm} represents the data after normalization.

During the training process, the model uses Adam to optimize the network parameters, with the learning rate is 0.001, the $\beta 1$ and $\beta 2$ are 0.9 and 0.999 respectively. Meanwhile, through the ReduceLROnPlateau learning rate reduction strategy, when the MAE of the validation set does not decrease for 5 consecutive epochs, the learning rate will be reduced by factor of 0.5. And the early stop mechanism is also introduced in the experiment to avoid overfitting.

The performance of the proposed network is evaluated under mean absolute error (MAE) and root mean square error (RMSE) criteria. Furthermore, we also calculated the error at each moment and evaluated the performance by the maximum error (Max Error). The MAE demonstrates the accuracy of the estimation while the RMSE demonstrates the robustness of the estimation. And the reliability of the prediction can be demonstrated by the Max Error.

The formulas of the three evaluation criteria mentioned above are as follows:

$$MAE = \frac{1}{n}\sum_{i=1}^{n}\left|y_i - \hat{y}_i\right| \tag{3}$$

$$RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2} \tag{4}$$

$$Maxerror = max\left|y_i - \hat{y}_i\right| \tag{5}$$

B. EXPERMENT DATASET

The Panasonic 18650PF battery dataset [36] was used for training and testing of the model, Panasonic 18650PF is a lithium nickel cobalt aluminum oxide chemistry battery. The details are shown in the Table 1.

To further validate the model to estimate performance under realistic conditions, it is necessary to provide a dataset collected in a wide range of practical driving environments to the model. In the experiment, we used nine driving cycles collected under three temperature conditions of 25°, 10°, and 0° including US06 Supplemental FTP Driving Schedule (US06), Highway Fuel Economy Driving Schedule (HWFET), Urban Dynamometer Driving Schedule (UDDS), and LA92 Dynamometer Driving Schedule (LA92) four

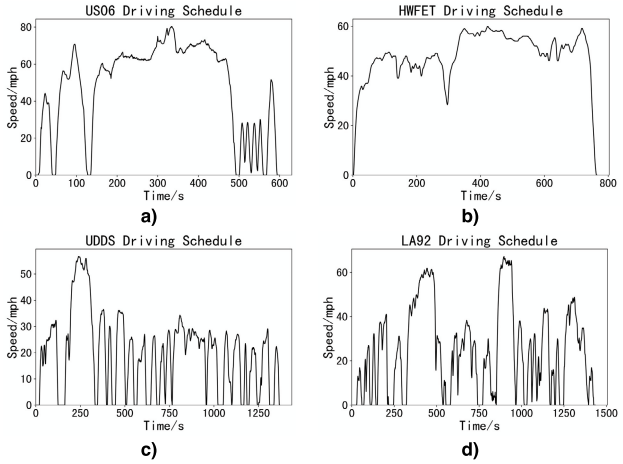


FIGURE 7. Dynamometer drive schedules. a) US06, b) HWFET, c) UDDS, d) LA92.

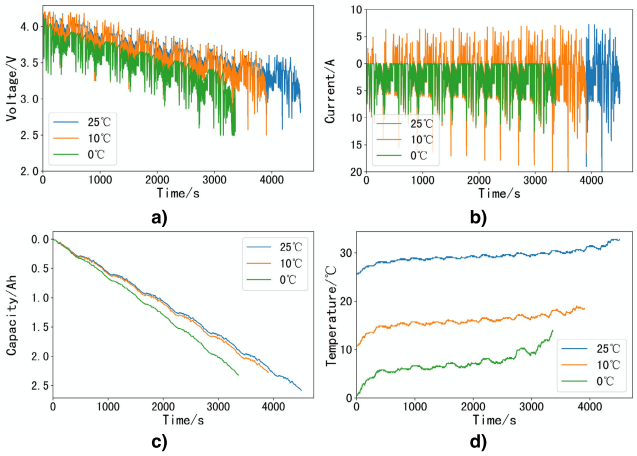


FIGURE 8. US06 drive cycle recorded at ambient temperature of 25°C, 10°C, and 0 °C. a) Voltage, b) Current, c) Capacity, d) Temperature.

common driving cycles, and Cycle 1, Cycle 2, Cycle 3, Cycle 4 four hybrid driving cycles, each of which is composed of US06, HWFET, UDDS, LA92 randomly mixed. Besides, a Neural Network driving cycle (Cycle NN) was also included, which can provide some additional dynamics data by combining portions of US06 and LA92 driving cycles specially. Fig. 7 shows the four common Dynamometer Drive Schedules described above, which are provided by the United States Environmental Protection Agency (EPA) [37].

The drive cycle power profile is calculated for an electric Ford F150 truck with a 35kWh battery pack scaled for a single 18650PF cell. The data collected in the data set includes voltage, current, amp-hours, and battery surface temperature. Fig. 8 shows the relevant measurement data of lithium-ion battery in US06 driving cycle collected under three temperature conditions including 25°C, 10°C, and 0°C. It is not difficult to find that when the battery is running in a low-temperature environment, the voltage drops faster, and the battery has no regenerative braking energy in a 0°C environment, resulting in a greatly shortened running time. It is a challenge for estimating SOC.

C. ESTIMATION AT FIXED AMBIENT TEMPERATURE

In this section, we used data sets collected at 25°C, 10°C, and 0°C to train and verify the model respectively. At three temperatures, data including US06, HWFET, UDDS, LA92, and Cycle NN five drive cycles were fed into the model for training, and the data including Cycle 1, Cycle 2, Cycle 3, and Cycle 4 mixed drive cycles at corresponding temperatures were used to verify the estimated performance of the model respectively.

The estimation curve of the model after training with single temperature data is shown in Fig. 9, including the estimated curve and error curve of the model at each moment. MAE, RMSE, and Max Error of model estimation results under three temperature conditions are listed in Table 2. The average MAE and average RMSE of SOC estimation results under 25°C, 10°C, and 0°C are 0.54%, 0.64%, and 1.29%, 0.78%, 0.81%, and 1.73%, respectively, and the maximum errors are 6.19%, 5.37%, and 7.68%, respectively. Which proved that the TCN model directly maps to SOC through the measured values of the battery under single temperature condition achieves satisfactory performance.

The results show that the proposed TCN model can capture the nonlinear relationship between lithium-ion battery measurement parameters and SOC by learning from the training data, and can estimate the corresponding SOC accurately using three measurement parameters including voltage, current, and temperature.

D. ESTIMATION AT MULTIPLE AMBIENT TEMPERATURES

The operation characteristics of lithium-ion batteries are different at different temperatures. For example, cells at 0° have no regenerative braking energy. The models in the previous section which trained under the condition of a single temperature cannot be well applied to other temperatures, which makes it necessary to train a model for each temperature. When applying this method to reality, training multiple models for different temperatures will not only increase the cost of this method but also complicate the application of this method.

Therefore, in this section, we use more comprehensive training data to train the TCN model. This training set contains a total of 15 drive cycles of US06, HWFET, UDDS, LA92, and Cycle NN, which are collected at 25°C, 10°C, and 0°C respectively. Different from the previous section, in this section, we hope to capture the non-linear relationship between battery measurement parameters and SOC values under different temperatures and working conditions through a single TCN model and estimate the SOC through this single model. We used data including Cycle 1, Cycle 2, Cycle 3, and Cycle 4 hybrid drive cycles collected at the above three temperatures to test the model performance.

The estimated performance of the trained TCN model at the above three temperatures is shown in Fig. 10. The results show that in the temperature environment of 25°C, 10°C, and 0°C, the estimated average MAE and average RMSE under the three temperatures are 0.67% and 0.87% respectively,

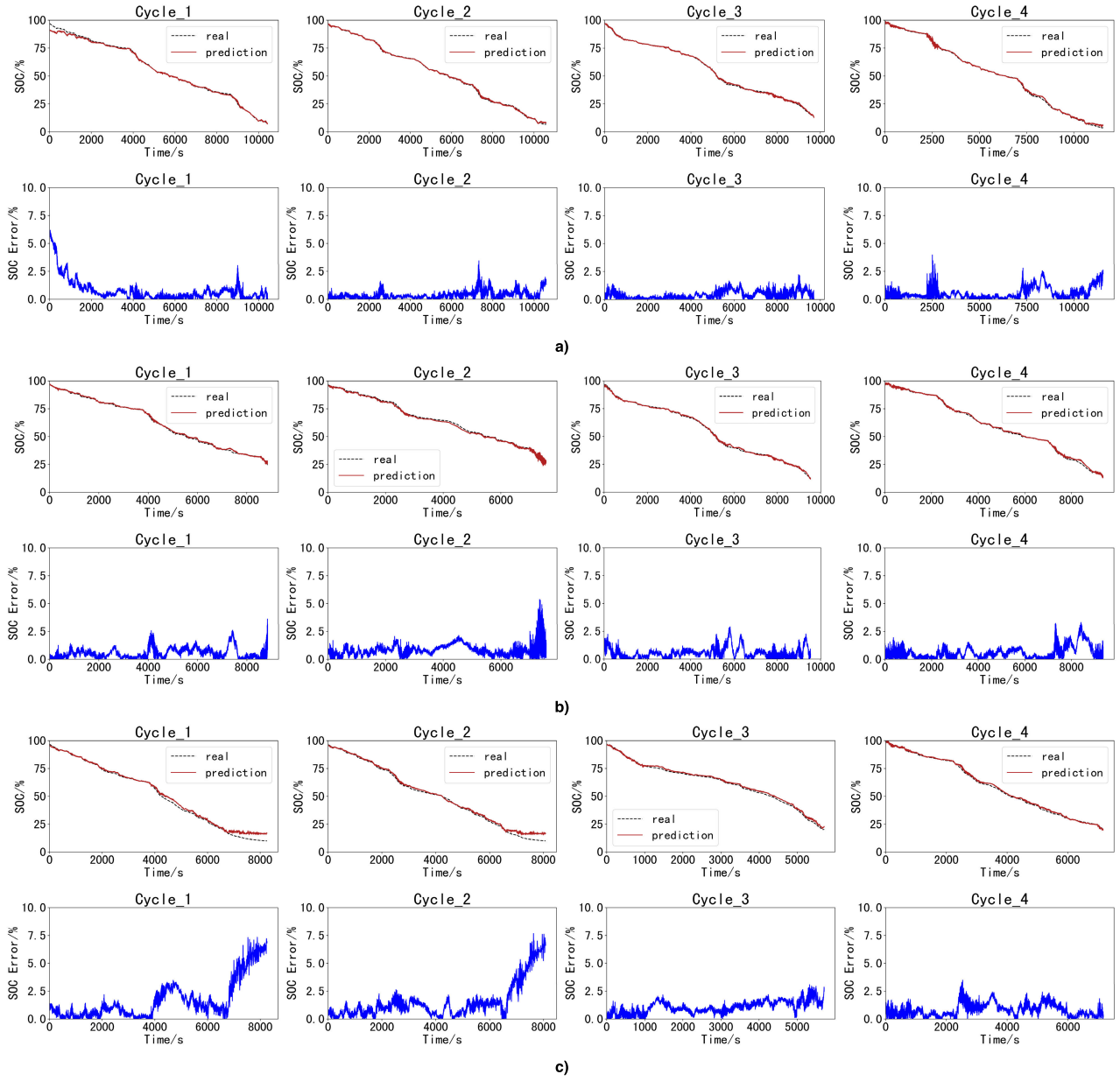


FIGURE 9. SOC estimation curves trained on fixed ambient temperature data: a) 25 °C, b) 10 °C, c) 0 °C.

TABLE 2. Estimation results at fixed ambient temperature.

Temperature	Condition	MAE (%)	RMSE (%)	Max Error (%)
25°C	Cycle1	0.78	1.27	6.19
	Cycle2	0.40	0.54	3.44
	Cycle3	0.41	0.54	2.23
	Cycle4	0.55	0.77	3.97
10°C	Cycle1	0.55	0.72	3.61
	Cycle2	0.82	0.95	5.37
	Cycle3	0.57	0.74	2.90
	Cycle4	0.61	0.83	3.31
0°C	Cycle1	1.71	2.48	7.34
	Cycle2	1.57	2.24	7.68
	Cycle3	1.07	1.21	3.08
	Cycle4	0.81	0.81	3.48

and the maximum error is 4.22%. Table 3 lists the detailed evaluation indexes of each test.

Experiments show that TCN can learn the relationship between battery measurable parameters and SOC at different temperatures with a single set of network parameters and can achieve SOC estimation in various working conditions through a single model. This greatly simplifies the complexity of the method, making it more convenient to apply in practical work.

E. ESTIMATION AT VARYING AMBIENT TEMPERATURES

The temperature is constantly changing throughout the day. Due to the actual driving environment, the working conditions of lithium-ion batteries in electric vehicles may experience certain temperature changes. Therefore, an additional test was added. The test data used in this additional test was the drive cycles including US06, HWFET, UDDS, and

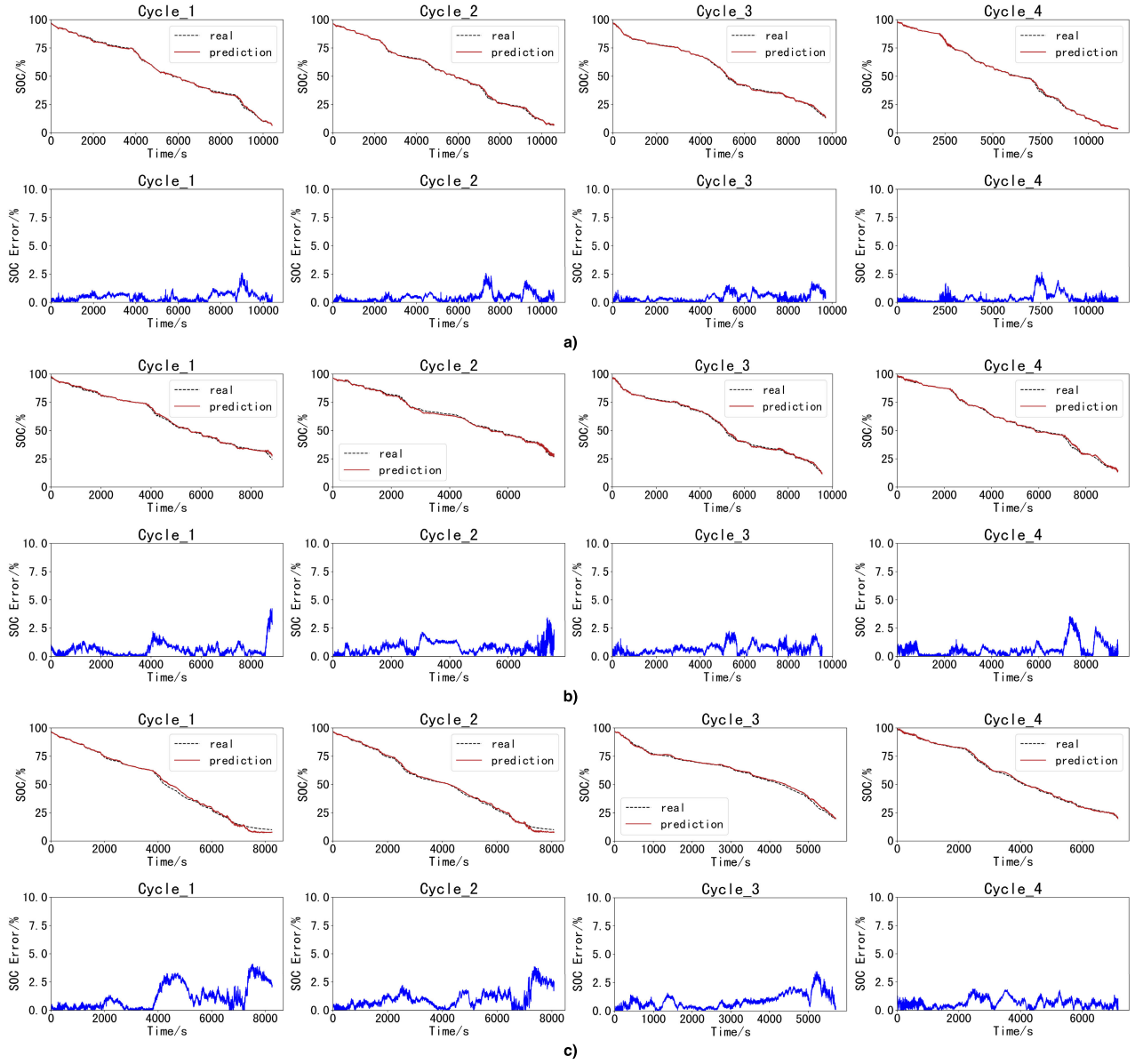


FIGURE 10. SOC estimation curves trained on multiple ambient temperature data: a) 25 °C, b) 10 °C, c) 0 °C.

TABLE 3. Estimation results at multiple ambient temperatures.

Temperature	Condition	MAE (%)	RMSE (%)	Max Error (%)
25°C	Cycle1	0.47	0.60	2.61
	Cycle2	0.44	0.60	2.57
	Cycle3	0.43	0.56	1.84
	Cycle4	0.36	0.57	2.67
10°C	Cycle1	0.59	0.82	4.22
	Cycle2	0.76	0.90	3.42
	Cycle3	0.67	0.80	2.23
	Cycle4	0.60	0.88	3.55
0°C	Cycle1	1.15	1.57	4.09
	Cycle2	1.02	1.28	3.86
	Cycle3	0.88	1.09	3.46
	Cycle4	0.64	0.76	1.88

LA92, which were collected under the condition of temperature rising from 10°C to 25°C, and in which, some of the temperatures do not exist in the previous training set. SOC estimation in such a rising ambient temperature environment is a test of generalization ability.

Fig. 11 shows the test results, including the estimated curve, error curve, and temperature change curve. Table 4 lists detailed evaluation indicators. In the varying temperature environment of 10°C to 25°C, the model estimates mean MAE and RMSE of 3.64% and 4.14% over the four drive cycles, respectively. Although there is a big gap with the results of the previous two experiments, the results are still acceptable, and the maximum error is controlled within 10%. It can be inferred that the model will have a more stable estimation of performance if the training set includes more abundant working condition information.

The results show that the proposed TCN model has a good generalization ability, and can correctly estimate SOC even

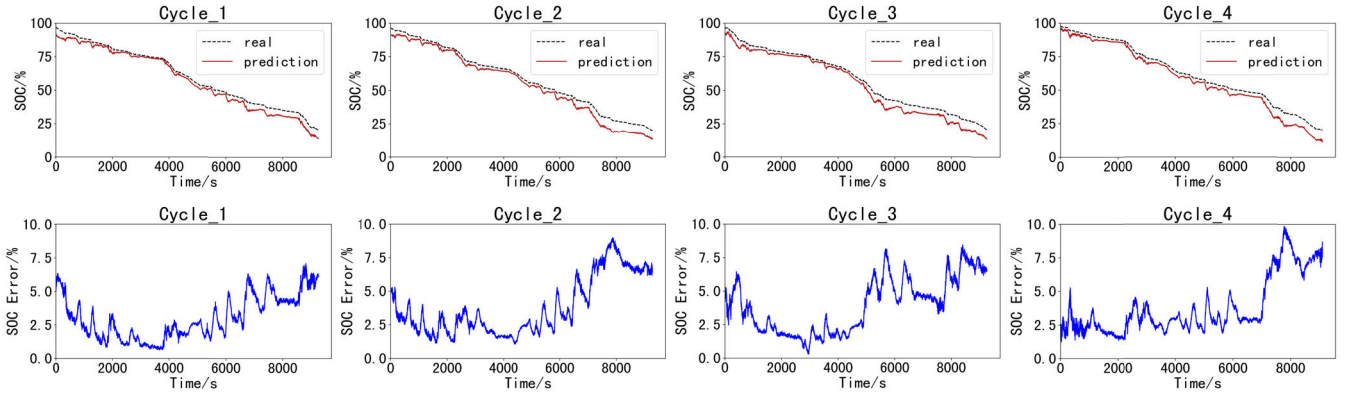


FIGURE 11. SOC estimation curves under rising ambient temperature.

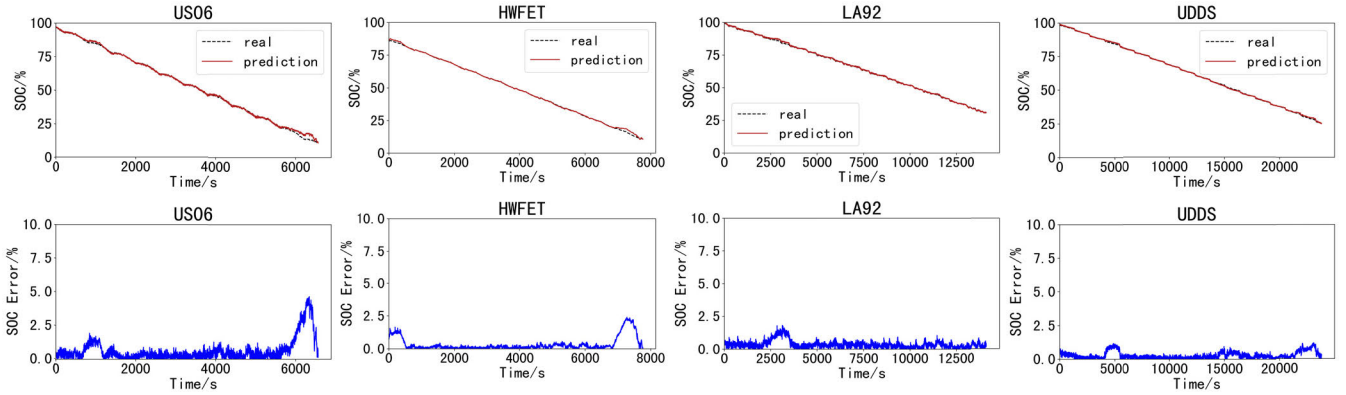


FIGURE 12. SOC estimation curves trained on fixed ambient temperature data of 25°C by transfer learning.

TABLE 4. Estimation results at rasing ambient temperature.

Temperature	Condition	MAE (%)	RMSE (%)	Max Error (%)
10°C-25°C	Cycle1	3.04	3.43	7.07
	Cycle2	3.80	4.36	9.00
	Cycle3	3.94	4.44	8.45
	Cycle4	3.79	4.34	9.87

in the absence of temperature in the training set. This is an important reference for extending the method to practical application.

F. ESTIMATION OF TRANSFER LEARNING MODEL

In this section, we use a small amount of data of Turnigy Graphene battery [38] to train the model combining transfer learning technology, so as to obtain a new model of SOC estimation that can be applied to this Turnigy Graphene battery by consuming less cost. Table 5 lists the main specification of the battery.

TCN model which is trained at a fixed ambient temperature of 25°C is selected as the pre-training model, and data from two mixed drive cycles of a Turnigy Graphene battery collected at 25°C is fed into it for transfer learning. Finally, we evaluated the model through the data of four cycles of driving including US06, HWFET, UDDS, and LA92 driving cycle. Its estimation curve and error curve are shown in

TABLE 5. Turnigy graphene battery main specifications.

Chemistry	LiPo
Nominal Voltage	3.7 V
Charge	4.2V, 50mA End-Current (CC-CV) Fast
Discharge	2.8V End Voltage, 20A MAX Continuous Current
Nominal Capacity	5 Ah
Energy Density	134 (Wh/Kg)

TABLE 6. Estimation results at 25°C ambient temperature.

Temperature	Condition	MAE (%)	RMSE (%)	Max Error (%)
25°C	US06	0.61	1.02	4.64
	HWFET	0.34	0.62	2.39
	UDDS	0.26	0.36	1.18
	LA92	0.35	0.45	1.84

Fig. 12, and the evaluation indexes of the results are listed in Table 6. The average MAE and average RMSE of the model estimate results are 0.39% and 0.61%, and the maximum error is 4.64%.

The results show that for a similar type of battery, only a small amount of data is needed to further train a model that has been trained on other battery data sets, and then the model can be migrated to estimate the SOC of another new battery through transfer learning. This reduces the cost of model acquisition and reduces the dependence of deep learning on a large amount of training data.

TABLE 7. Comparison of estimation results with other methods.

Method	Related Information	Error
LSTM-RNN [21]	Input: $[V(k), I(k), T(k)]$ Output: $[SOC(k)]$ Test case condition: 25C°, 10C°, and 0C°	MAE=1.21%, RMSE=1.52%, MAX=6.69%
GRU-RNN [24]		MAE=0.86%, MAX=7.59%
TCN (proposed method)		MAE=0.67%, RMSE=0.87%, MAX=4.22%

G. COMPARATIVE ANALYSIS OF EXPERIMENTAL RESULTS

To further prove the advantages of the TCN model, we collated the experimental results of an LSTM-RNN model and a GRU-RNN model that did under multi-temperature conditions. Same as part D of Section IV, both the two models used the data of multiple driving cycles collected under three temperature ambients of 25°C, 10°C, and 0°C for model training and testing. And then we compared them with the TCN model.

All these three models can use the measurements to estimate SOC directly under various working conditions through a single model. Details are shown in Table 7. MAE and RMSE are the average values of test results of various conditions, and MAX is the maximum error of all the tests. It shows that the TCN model proposed in this paper performs better than the two models in accuracy.

V. CONCLUSION

This paper describes how the TCN network maps voltage, current, and temperature directly to the SOC of the battery, and through its specially designed dilated causal convolution structure, it shows superior ability to process the timing data of lithium-ion battery. During the training process, the model can be updated with parameters through self-learning. The model after training has good accuracy, and the SOC of lithium-ion battery under different temperatures and working conditions can be estimated by just one model. Compared with the traditional method, this proposed method does not require the battery model establishment manually or determination and calculation of a large number of parameters, and the method of directly mapping battery measurement parameters to SOC is more suitable for the on-board system. At the same time, the performance of the model is extensively verified by a public data set, and the estimation effect of its generalization to a realistic situation is tested. Even in the absence of working conditions in the training set, the model still achieves good performance. In addition, this paper also verifies the ability of the proposed model to be applied to other types of lithium-ion battery SOC estimation through transfer learning technology. In conclusion, the TCN-based method proposed in this paper to estimate the SOC of lithium-ion batteries has been proved to be feasible, and it also plays a certain reference role in the treatment of other similar timing problems. However, considering that the computing capacity of the vehicle system is relatively weak, there are certain requirements for the estimated speed of SOC. It can be considered to introduce neural network compression technology into model training, then the network scale can be reduced by reducing parameters, so as to be more suitable for the vehicle system.

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YUEFENG LIU received the B.S. degree from the Inner Mongolia University of Technology, Huhhot, China, in 2001, the M.S. degree from the Inner Mongolia University of Science and Technology, Baotou, China, in 2008, and the Ph.D. degree from the Harbin Institute of Technology, Harbin, China, in 2020.

He has been an Associate Professor with the Inner Mongolia University of Science and Technology since 2011. His research interests include prognostics and health management, and deep leaning.



JIAQI LI received the B.S. degree from the Inner Mongolia University of Science and Technology, Baotou, China, in 2019, where he is currently pursuing the M.S. degree.

His research interests include prognostics and health management, and deep leaning.



GONG ZHANG received the B.S. degree from Shanxi Datong University, Shanxi, China, in 2018. He is currently pursuing the M.S. degree with the Inner Mongolia University of Science and Technology, Baotou, China.

His research interests include prognostics and health management, and deep leaning.



BIN HUA received the M.S. degree from Sichuan University, Sichuan, China.

He is currently an Experimental Technician with the School of Information Engineering, Ningxia University, China. During the work, participated in completing four national projects, three autonomous regional projects, and one horizontal project. In 2020 "Challenge Cup" Entrepreneurship Plan Competition for College Students in the District, won the title of Excellent Advisor.



NEAL XIONG (Senior Member, IEEE) received the Ph.D. degree in software engineering from Wuhan University, and the Ph.D. degree in dependable networks from the Japan Advanced Institute of Science and Technology.

He was a Full Professor with Colorado Technical University for a period of four years, an Assistant Professor with the Wentworth Technology Institution for a period of one year, and an Assistant Professor and held a postdoctoral position with Georgia State University for a period of four years. He is currently with the Department of Mathematics and Computer Science, Northeastern State University, Tahlequah, OK, USA. He has published over 100 international journal articles and over 100 international conference papers. Some of his works were published in the IEEE JOURNAL ON SELECTED AREAS IN COMMUNICATIONS, the IEEE or ACM Transactions, ACM Sigcomm Workshop, and the IEEE INFOCOM, ICDCS, and IPDPS. His research interests include cloud computing, business networks, security and dependability, parallel and distributed computing, and optimization theory. He has been a PC member, an OC member, a General Chair, a Program Chair, and a Publicity Chair of over 100 international conferences. He is also a Senior Member of the IEEE Computer Society. He received the Best Paper Award in the 10th IEEE International Conference on High Performance Computing and Communications 2008 and the Best Student Paper Award in the 28th North American Fuzzy Information Processing Society Annual Conference 2009. He is also the Chair of Trusted Cloud Computing Task Force, the IEEE Computational Intelligence Society, and the Industry System Applications Technical Committee. He was serving as the Editor-in-Chief, an Associate Editor, or an Editor Member for over ten international journals, including an Associate Editor for the IEEE TRANSACTIONS ON SYSTEMS, MAN AND CYBERNETICS SYSTEMS and *Information Science*, an Editor-in-Chief for the *Journal of Internet Technology* and the *Journal of Parallel and Cloud Computing*, and a Guest Editor for over ten international journals, including the IEEE WIRELESS COMMUNICATIONS, the IEEE SENSORS JOURNAL, *Wireless Networks (WINET)*, and *Mobile Networks and Applications (MONET)*.